

Extended Body Control HLD

Document ID: HLD-BCM-LARGE-001 **Version:** 1.0 **Status:** Synthetic Demonstration Document

Purpose: This synthetic document is intended only for scalability and retrieval demonstration of the AUTOSAR HLD Document Analysis Assistant. It is not an OEM production HLD.

1. Executive Overview

This High-Level Design describes a synthetic body-control architecture used to demonstrate long-document ingestion and semantic retrieval. It covers exterior lighting, door-state processing, interior lighting, diagnostics and gateway communication.

The architecture is organized around explicit software components, typed interfaces, provided/required ports, signals, and component dependencies. The document intentionally includes repeated engineering context around unique architecture facts so that semantic retrieval can be exercised on a longer document.

2. Architecture Summary

Component	Responsibility	Key Interface	Primary Dependency
LightSwitch	Captures driver exterior-light requests.	LightRequestInterface	No upstream software component; provides request.
BodyControlManager	Coordinates body-control functions and processes light requests.	BodyCommandInterface	Requires LightSwitch and DoorStateManager.
ExteriorLightController	Controls exterior lighting based on body commands.	LightStatusInterface	Requires BodyControlManager.
DoorStateManager	Aggregates door-state information for body functions.	DoorStateInterface	Requires DoorSensorAdapter.
DoorSensorAdapter	Adapts raw door sensor states into architecture-level information.	DoorSensorInterface	No upstream software component.
InteriorLightController	Controls interior lighting using door state and body commands.	InteriorLightStatusInterface	Requires DoorStateManager and BodyControlManager.
BodyDiagnostics	Monitors body-control components and publishes diagnostic data.	BodyDiagnosticInterface	Requires BodyControlManager and ExteriorLightController.
BodyGateway	Exposes selected body status to external vehicle domain.	BodyGatewayInterface	Requires BodyControlManager and BodyDiagnostics.

3. Component Design - LightSwitch

Responsibility. Captures driver exterior-light requests. The component participates in the body control control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **LightRequestInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. LightSwitch has the following primary dependency context: No upstream software component; provides light request.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
LightSwitchOut	P-Port	LightRequestInterface	Provides LightRequestInterface data where applicable.
LightSwitchIn	R-Port	SwitchHealthInterface	Consumes SwitchHealthInterface input where applicable.

3.1 Engineering Notes

LightSwitch is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that LightRequestInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement

without sending the full document to the language model.

3.2 Engineering Notes

LightSwitch is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that LightRequestInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

3.3 Engineering Notes

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Review focus 3: verify that LightRequestInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

3.4 Engineering Notes

LightSwitch is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 4: verify that LightRequestInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

4. Component Design - BodyControlManager

Responsibility. Coordinates body-control functions and processes light requests. The component participates in the body control control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **BodyCommandInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. BodyControlManager has the following primary dependency context: Requires LightSwitch and DoorStateManager.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
BodyControlManagerOut	P-Port	BodyCommandInterface	Provides BodyCommandInterface data where applicable.
BodyControlManagerIn	R-Port	LightRequestInterface	Consumes LightRequestInterface input where applicable.

4.1 Engineering Notes

BodyControlManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that BodyCommandInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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BodyControlManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that BodyCommandInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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4.4 Engineering Notes

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Review focus 4: verify that BodyCommandInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

5. Component Design - ExteriorLightController

Responsibility. Controls exterior lighting based on body commands. The component participates in the body control control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **LightStatusInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. ExteriorLightController has the following primary dependency context: Requires BodyControlManager.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
ExteriorLightControllerOut	P-Port	LightStatusInterface	Provides LightStatusInterface data where applicable.
ExteriorLightControllerIn	R-Port	BodyCommandInterface	Consumes BodyCommandInterface input where applicable.

5.1 Engineering Notes

ExteriorLightController is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that LightStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

5.2 Engineering Notes

ExteriorLightController is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that LightStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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ExteriorLightController is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 4: verify that LightStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement

without sending the full document to the language model.

6. Component Design - DoorStateManager

Responsibility. Aggregates door-state information for body functions. The component participates in the body control control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **DoorStateInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. DoorStateManager has the following primary dependency context: Requires DoorSensorAdapter.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
DoorStateManagerOut	P-Port	DoorStateInterface	Provides DoorStateInterface data where applicable.
DoorStateManagerIn	R-Port	DoorSensorInterface	Consumes DoorSensorInterface input where applicable.

6.1 Engineering Notes

DoorStateManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that DoorStateInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

6.2 Engineering Notes

DoorStateManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that DoorStateInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

6.3 Engineering Notes

DoorStateManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

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6.4 Engineering Notes

DoorStateManager is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 4: verify that DoorStateInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

7. Component Design - DoorSensorAdapter

Responsibility. Adapts raw door sensor states into architecture-level information. The component participates in the body control control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **DoorSensorInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. DoorSensorAdapter has the following primary dependency context: No upstream software component.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
DoorSensorAdapterOut	P-Port	DoorSensorInterface	Provides DoorSensorInterface data where applicable.
DoorSensorAdapterIn	R-Port	SensorHealthInterface	Consumes SensorHealthInterface input where applicable.

7.1 Engineering Notes

DoorSensorAdapter is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that DoorSensorInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

7.2 Engineering Notes

DoorSensorAdapter is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that DoorSensorInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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DoorSensorAdapter is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

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DoorSensorAdapter is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 4: verify that DoorSensorInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement

without sending the full document to the language model.

8. Component Design - InteriorLightController

Responsibility. Controls interior lighting using door state and body commands. The component participates in the body control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **InteriorLightStatusInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. InteriorLightController has the following primary dependency context: Requires DoorStateManager and BodyControlManager.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
InteriorLightControllerOut	P-Port	InteriorLightStatusInterface	Provides InteriorLightStatusInterface data where applicable.
InteriorLightControllerIn	R-Port	DoorStateInterface	Consumes DoorStateInterface input where applicable.

8.1 Engineering Notes

InteriorLightController is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that InteriorLightStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

8.2 Engineering Notes

InteriorLightController is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that InteriorLightStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

8.3 Engineering Notes

InteriorLightController is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 3: verify that InteriorLightStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

8.4 Engineering Notes

InteriorLightController is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 4: verify that InteriorLightStatusInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

9. Component Design - BodyDiagnostics

Responsibility. Monitors body-control components and publishes diagnostic state. The component participates in the body control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **BodyDiagnosticInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. BodyDiagnostics has the following primary dependency context: Requires BodyControlManager and ExteriorLightController.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
BodyDiagnosticsOut	P-Port	BodyDiagnosticInterface	Provides BodyDiagnosticInterface data where applicable.
BodyDiagnosticsIn	R-Port	LightStatusInterface	Consumes LightStatusInterface input where applicable.

9.1 Engineering Notes

BodyDiagnostics is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that BodyDiagnosticInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

9.2 Engineering Notes

BodyDiagnostics is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that BodyDiagnosticInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

9.3 Engineering Notes

BodyDiagnostics is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 3: verify that BodyDiagnosticInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

9.4 Engineering Notes

BodyDiagnostics is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 4: verify that BodyDiagnosticInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model.

This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

10. Component Design - BodyGateway

Responsibility. Exposes selected body status to external vehicle domains. The component participates in the body control architecture and exchanges data through explicitly defined AUTOSAR-style interfaces.

Interface behavior. The principal interface for this component is **BodyGatewayInterface**. Its architecture relationship is documented so that provider/consumer consistency can be checked automatically.

Dependency. BodyGateway has the following primary dependency context: Requires BodyControlManager and BodyDiagnostics.. Dependency information is used by impact analysis, functional-flow construction and architecture graph generation.

Port Name	Type	Interface	Description
BodyGatewayOut	P-Port	BodyGatewayInterface	Provides BodyGatewayInterface data where applicable.
BodyGatewayIn	R-Port	BodyDiagnosticInterface	Consumes BodyDiagnosticInterface input where applicable.

10.1 Engineering Notes

BodyGateway is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 1: verify that BodyGatewayInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

10.2 Engineering Notes

BodyGateway is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 2: verify that BodyGatewayInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

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BodyGateway is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

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BodyGateway is reviewed with respect to interface ownership, data-flow direction, dependency traceability, initialization assumptions, diagnostic visibility and change impact. In a production program these sections would normally include project-specific requirements, timing constraints, safety considerations and interface contracts. For this synthetic scalability document, the text provides realistic surrounding context while keeping the architecture facts deterministic and reviewable.

Review focus 4: verify that BodyGatewayInterface remains traceable to the component responsibility, that required and provided ports are consistent, and that downstream consumers can be identified through the dependency model. This context is intentionally verbose enough to test whether semantic retrieval can locate the relevant architecture statement without sending the full document to the language model.

11. Signal Definitions

Signal	Interface	Unit	Description
LightRequest	LightRequestInterface	enum	Requested exterior-light mode.
BodyCommand	BodyCommandInterface	enum	Body-control command state.
LightStatus	LightStatusInterface	enum	Exterior-light operating status.
DoorState	DoorStateInterface	enum	Aggregated door-open/closed state.
BodyDiagnosticStatus	BodyDiagnosticInterface	enum	Body-domain diagnostic state.

12. Functional Flows

Flow 1: LightSwitch -> BodyControlManager -> ExteriorLightController

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

Flow 2: DoorSensorAdapter -> DoorStateManager -> InteriorLightController

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

Flow 3: BodyControlManager -> BodyDiagnostics -> BodyGateway

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

Flow 4: DoorStateManager -> BodyControlManager

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

Flow 5: BodyControlManager -> InteriorLightController

The flow is expressed at component level for architecture review. It can be used to reason about upstream providers, downstream consumers, change impact and test-planning scope.

13. Interface Governance

Interface Governance consideration 1. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

Interface Governance consideration 2. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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Interface Governance consideration 4. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

Interface Governance consideration 5. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that

architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

Interface Governance consideration 6. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

14. Diagnostics and Observability

Diagnostics and Observability consideration 1. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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15. Timing and Data Freshness

Timing and Data Freshness consideration 1. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

Timing and Data Freshness consideration 2. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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16. Fault Handling

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17. Change Impact Analysis

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18. Verification Considerations

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19. Deployment Assumptions

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20. Traceability Notes

Traceability Notes consideration 1. The body control architecture shall preserve explicit ownership of interfaces and traceability between component responsibilities, ports, signals and dependencies. Reviewers should verify that architecture changes remain visible in revision comparison and that document-specific questions can be supported by retrieved evidence. This synthetic paragraph represents the type of contextual engineering prose that can surround structured facts in a longer HLD.

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Appendix A - Architecture Inventory

LightSwitch: responsibility=Captures driver exterior-light requests.; interface=LightRequestInterface; dependency=No upstream software component; provides light request..

BodyControlManager: responsibility=Coordinates body-control functions and processes light requests.; interface=BodyCommandInterface; dependency=Requires LightSwitch and DoorStateManager..

ExteriorLightController: responsibility=Controls exterior lighting based on body commands.; interface=LightStatusInterface; dependency=Requires BodyControlManager..

DoorStateManager: responsibility=Aggregates door-state information for body functions.; interface=DoorStateInterface; dependency=Requires DoorSensorAdapter..

DoorSensorAdapter: responsibility=Adapts raw door sensor states into architecture-level information.; interface=DoorSensorInterface; dependency=No upstream software component..

InteriorLightController: responsibility=Controls interior lighting using door state and body commands.; interface=InteriorLightStatusInterface; dependency=Requires DoorStateManager and BodyControlManager..

BodyDiagnostics: responsibility=Monitors body-control components and publishes diagnostic state.; interface=BodyDiagnosticInterface; dependency=Requires BodyControlManager and ExteriorLightController..

BodyGateway: responsibility=Exposes selected body status to external vehicle domains.; interface=BodyGatewayInterface; dependency=Requires BodyControlManager and BodyDiagnostics..

Appendix B - Scalability Test Notes

This document is deliberately longer than the controlled accuracy benchmark. It should be used to demonstrate ingestion, chunk creation, vector indexing and retrieval behavior. It should not be merged into the existing precision/recall/F1 benchmark unless a separate page-level/entity-level ground truth and evaluation protocol are defined.